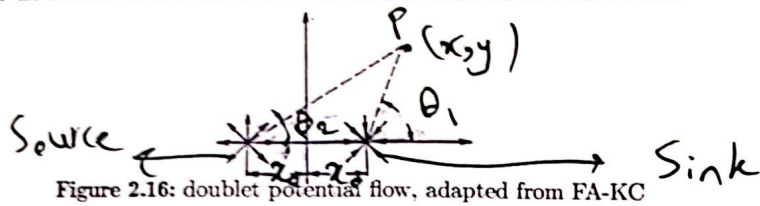


Doublet flow - doublet flow

are going to use the principle of superposition to derive our third - a doublet. Consider a source and a sink as shown below.

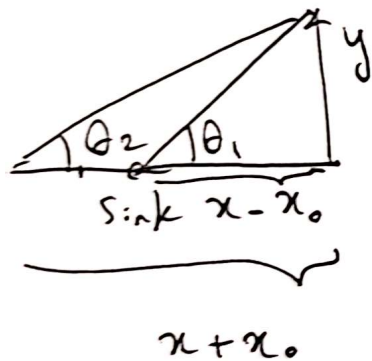


stream function for the entire flow field is now

$$\psi = \psi_{\text{source}} + \psi_{\text{sink}} \quad (2.89)$$

$$\psi = -\frac{K}{2\pi}(\theta_1 - \theta_2) = -\frac{K}{2\pi} \left(\tan^{-1} \frac{y}{x-x_0} - \tan^{-1} \frac{y}{x+x_0} \right) \quad (2.90)$$

©, EPG Pot. Flow, Elim. Flows, 85



$$\nabla^2 \phi = \nabla^2 \psi = 0$$

Potential flow

Doublet
Vortex

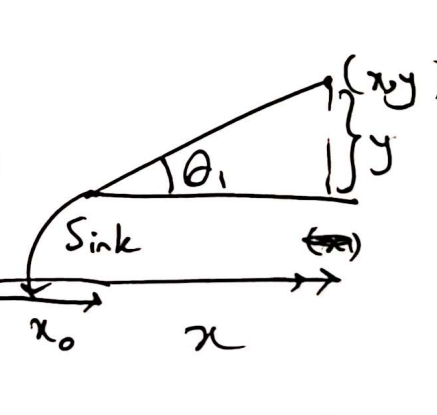
non-lifting cylinder

lifting cylinder

Examples

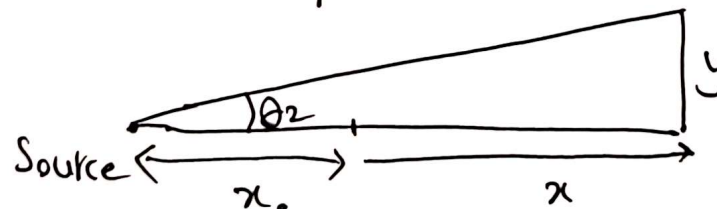
We discussed

Uniform flow
Source/Sink



$$\theta_1 = \tan^{-1} \frac{y}{x-x_0}$$

$$\theta_2 = \tan^{-1} \frac{y}{x+x_0}$$



ved flow - doublet flow

$$\tan^{-1} \alpha - \tan^{-1} \beta = \tan^{-1} \left(\frac{\alpha - \beta}{1 + \alpha\beta} \right) \quad (2.91)$$

After some algebra, $\psi(x, y) = -\frac{K}{2\pi} \tan^{-1} \frac{2x_0 y}{x^2 + y^2 - x_0^2} \quad (2.92)$

let, $B = 2x_0 K \rightarrow$ doublet strength when $x_0 \rightarrow 0 \quad (2.93)$

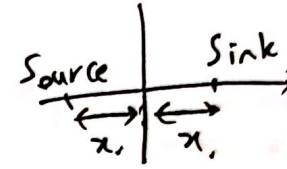
therefore, $\psi_{doublet} = \lim_{x_0 \rightarrow 0} \left[-\frac{B}{4\pi x_0} \tan^{-1} \left(\frac{2x_0 y}{x^2 + y^2 - x_0^2} \right) \right] \quad (2.94)$

$x_0 \rightarrow 0, \frac{2x_0 y}{x^2 + y^2 - x_0^2} \rightarrow$ something small (2.95)

therefore as $x_0 \rightarrow 0, \tan^{-1} \left(\frac{2x_0 y}{x^2 + y^2 - x_0^2} \right) \rightarrow \frac{2x_0 y}{x^2 + y^2 - x_0^2} \quad (2.96)$

$$B = 2x_0 K$$

$$\frac{K}{2\pi} = \frac{B}{4\pi x_0}$$



$$x_0 \rightarrow 0$$

$$\tan^{-1}(\epsilon) \sim \epsilon$$

$$\epsilon \rightarrow 0$$

$$\text{therefore, } \psi_{\text{doublet}} = \lim_{x_0 \rightarrow 0} \left[-\frac{B}{4\pi x_0} \frac{2x_0 y}{x^2 + y^2 - x_0^2} \right] \rightarrow 0 \quad (2.97)$$

$$= -\frac{B}{2\pi} \frac{y}{x^2 + y^2} = -\frac{B \sin \theta}{2\pi r} \quad (2.98)$$

$$v_r = -\frac{B \cos \theta}{2\pi r^2} \quad v_\theta = -\frac{B \sin \theta}{2\pi r^2} \quad (2.99)$$

$$\phi = \frac{B \cos \theta}{2\pi r} \quad \psi = -\frac{B \sin \theta}{2\pi r} \quad (2.100)$$

Even though we derived this flow, now that we have the final it i.e., ψ and ϕ , the doublet also becomes an elementary flow. We just use it, by adding it to other flows. This is the power of linear els!

Steady
flow
 $\nabla \cdot \mathbf{V} = 0$
 $\nabla \times \mathbf{V} = 0$
 $\mathbf{V} = \nabla \phi$
 $\nabla \times \nabla \phi = 0$

$$V_x = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}$$

$$V_y = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$\psi_{\text{doublet}} = \frac{-B}{2\pi} \frac{y}{x^2 + y^2} = \frac{-B}{2\pi} \frac{\sin \theta}{r}$$

have only
 $\underline{v_r}$

have
both
 $\begin{cases} v_r \\ v_\theta \end{cases}$

$$\psi_{\text{source}} = \frac{k}{2\pi} \theta, \quad \psi_{\text{sink}} = -\frac{k}{2\pi} \theta$$

$$\psi_{\text{doublet}} = \frac{-B}{2\pi} \frac{\sin \theta}{r}$$

$$V_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{\partial \phi}{\partial r}$$

$$V_\theta = -\frac{\partial \psi}{\partial r} = \frac{1}{r} \frac{\partial \phi}{\partial \theta}$$

Double check

irved flow - doublet flow

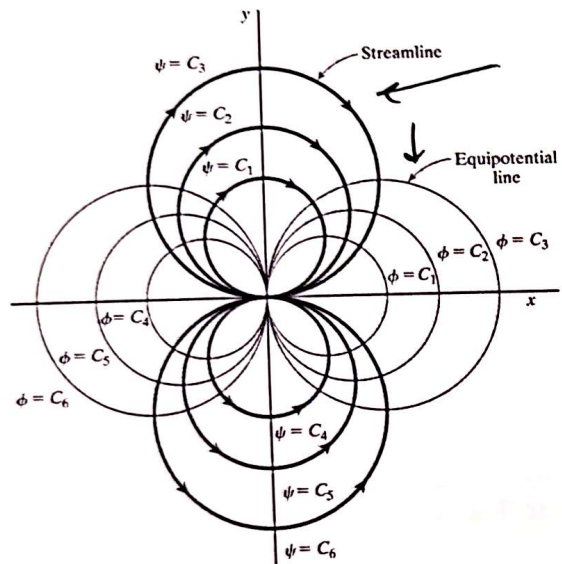
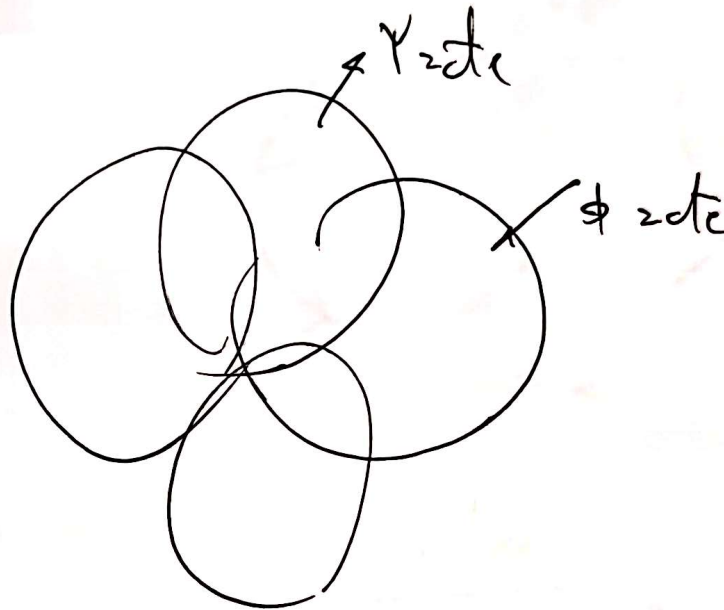


Figure 2.17: Fig. 3.11 from AE-BC

19, EPG Pot. Flow, Elim. Flows, 88



Vortex flow

$$\phi = \text{const} \rightarrow \psi = \text{const}$$

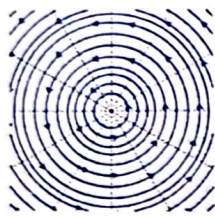
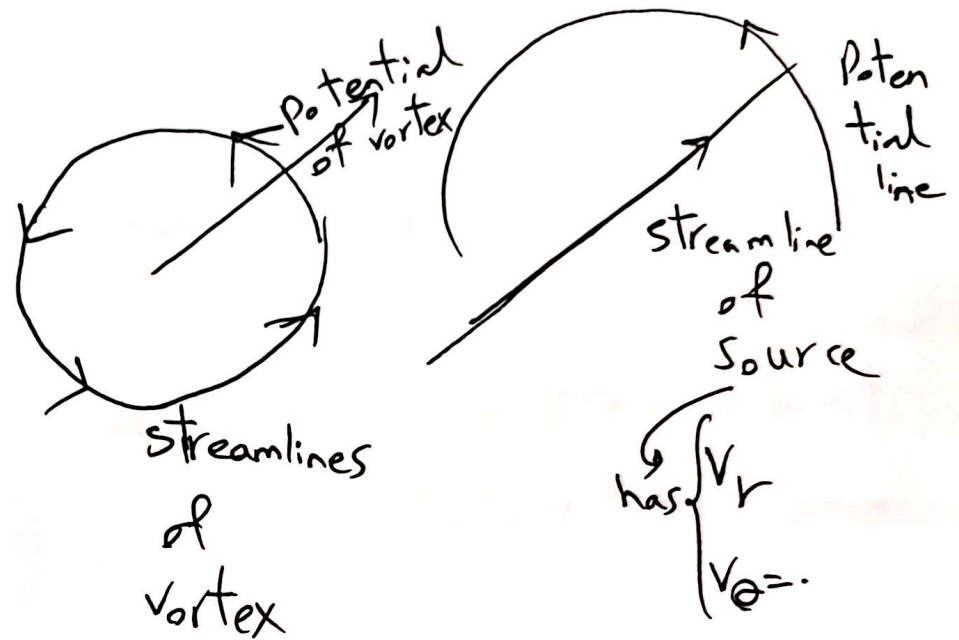


Figure 2.18: Vortex potential flow

$$v_r = 0 \quad v_\theta = -\frac{\Gamma}{2\pi r} \quad (2.101)$$

$$\phi = -\frac{\Gamma}{2\pi} \theta \quad \psi = \frac{\Gamma}{2\pi} \ln r \quad (2.102)$$

θ is in radians, as in the source and the sink Γ is centered at the origin.



has $\left\{ \begin{aligned} V_\theta &= -\frac{\Gamma}{2\pi r} = -\frac{\partial \psi}{\partial r} = -\frac{1}{r} \frac{\partial \phi}{\partial \theta} \\ V_r &= 0 \end{aligned} \right.$

$$\psi = \frac{\Gamma}{2\pi} \ln r$$

$$\phi = -\frac{\Gamma}{2\pi} \theta$$

mary

TABLE 3.3 Stream Functions and Potential Functions for Elementary Flows

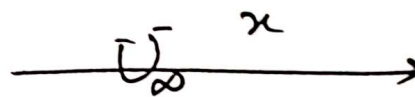
Flow	ψ	ϕ
Uniform flow in x direction	$U_\infty y$	$U_\infty x$
Uniform flow at angle α	$U_\infty (y \cos \alpha - x \sin \alpha)$	$U_\infty (x \cos \alpha + y \sin \alpha)$
Source	$\frac{K\theta}{2\pi}$	$\frac{K}{2\pi} \ln r$
Doublet	$-\frac{B}{2\pi} \sin \theta$	$\frac{B}{2\pi} \cos \theta$
Vortex (with clockwise circulation)	$-\frac{\Gamma}{2\pi} \ln r$	$\frac{\Gamma\theta}{2\pi}$
90° corner flow	Axy	$\frac{1}{2}A(x^2 - y^2)$
Solid-body rotation	$\frac{1}{2}\omega r^2$	Does not exist since flow is rotational

Figure 2.19: Table 3.3 from AE-BC

EPG Pot. Flow, Elem. Flows, 90

Navigation icons

$$\frac{1}{2}\omega r^2$$



$$u = \frac{\partial \psi}{\partial y} = U_\infty \Rightarrow$$

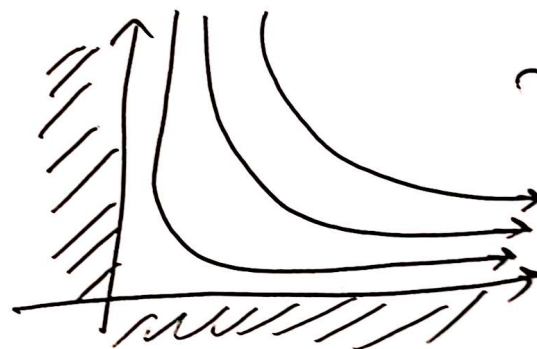
$$\psi = U_\infty y$$

$$\phi = U_\infty x$$



$$\psi = U_\infty (y \cos \alpha - x \sin \alpha)$$

$$\phi = U_\infty (x \cos \alpha + y \sin \alpha)$$

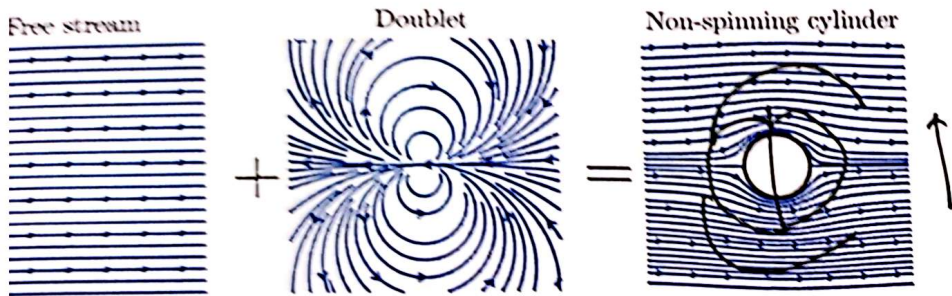


$$\psi = Axy$$

$$\phi = \frac{1}{2}A(x^2 - y^2)$$

AERODYNAMIC FORCES AND MOMENTS

Non-spinning cylinder (1st Sec. 3.13)



2.20: Potential flow over a non-spinning cylinder. Based on image from FA-JA

$$\psi(r, \theta) = \psi_{\text{free-stream}} + \psi_{\text{doublet}} \quad (2.103)$$

$$= U_{\infty} r \sin \theta + \frac{-B \sin \theta}{2\pi r} \quad (2.104)$$

$$\psi = U_{\infty} r \sin \theta \left(1 - \frac{B}{2\pi U_{\infty} r^2} \right) \quad (2.105)$$

$$\begin{aligned} \chi_{u_i} &= U_{\infty} y \\ \chi_{u_i} &= U_{\infty} r \sin \theta \end{aligned}$$

AERODYNAMIC FORCES AND MOMENTS

$y \rightarrow n$

potential flow over a non-spinning cylinder

$$\psi = U_{\infty} r \sin \theta \left(1 - \frac{B}{2\pi U_{\infty} r^2} \right) \quad (2.106)$$

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = U_{\infty} \cos \theta \left[1 - \frac{B}{2\pi U_{\infty} r^2} \right] \quad (2.107)$$

$$v_{\theta} = -\frac{\partial \psi}{\partial r} = -U_{\infty} \sin \theta \left[1 + \frac{B}{2\pi U_{\infty} r^2} \right] \quad (2.108)$$

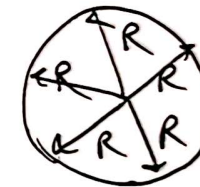
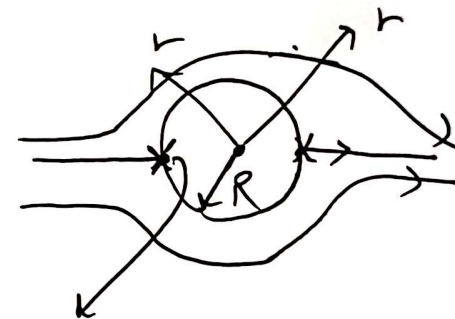
Stagnation pt. $|\vec{V}| = 0 \rightarrow r = R = \sqrt{\frac{B}{2\pi U_{\infty}}} \quad (2.109)$

$$C_p = 1 - \left(\frac{U}{U_{\infty}} \right)^2 = 1 - \left(\frac{\sqrt{v_r^2 + v_{\theta}^2}}{U_{\infty}} \right)^2 \quad (2.110)$$

$$C_p = 1 - 4 \sin^2 \theta \quad (2.111)$$

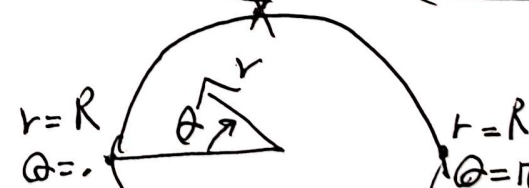
$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}$$

$$v_{\theta} = -\frac{\partial \psi}{\partial r}$$



stagnation point ($v_r = 0$, $v_{\theta} = 0$)

$$\theta = \pi/2 \quad v_{\theta} = -2U_{\infty}$$



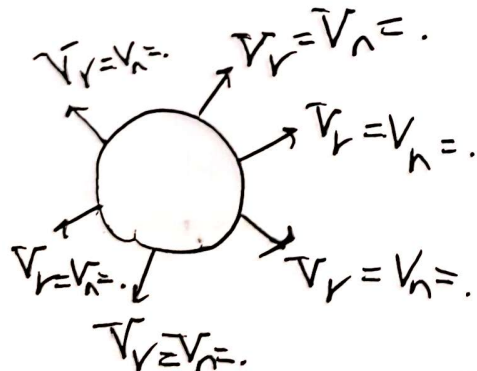
$$v_{\theta} = 2U_{\infty}$$

$\theta, \pi, 2\pi, \dots$

$$v_r = 0 \rightarrow \begin{cases} 1 = \frac{B}{2\pi U_{\infty} r^2} \\ r = \sqrt{\frac{B}{2\pi U_{\infty}}} \\ r = R \end{cases}$$

$2 \neq 0$

$1 = \sin \theta = 0$



potential flow over a non-spinning cylinder

defined on last page

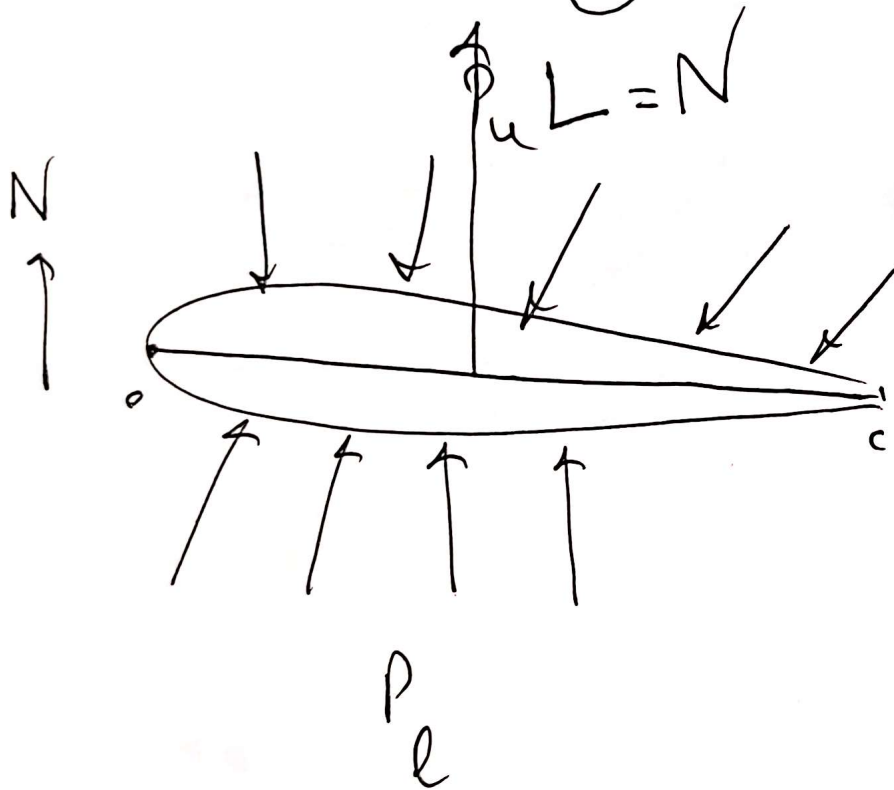
$$\psi = U_{\infty} r \sin \theta \left[1 - \left(\frac{R}{r} \right)^2 \right] \quad (2.112)$$

$$\psi(r = R) = 0 \rightarrow \text{surface streamline} \quad (2.113)$$

i.e., that it just happens that $\psi = 0$ is the surface streamline. There is no other significance to this.

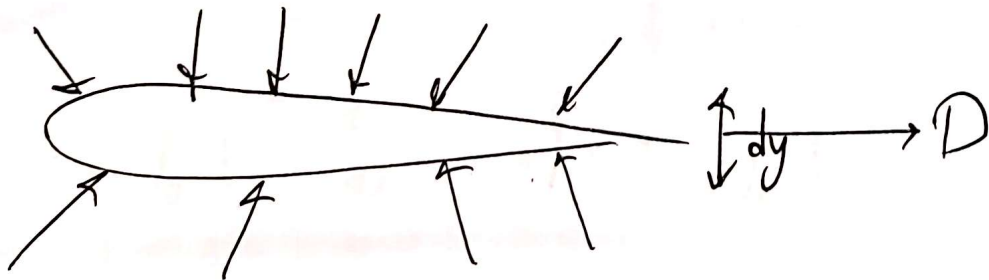
- Stagnation point $\rightarrow \theta = 0$ & π , $r = R$. (Show as an ungraded homework)
- Max velocity $\rightarrow r = R$, $\theta = \pi/2$ & $3\pi/2$, $v_r = 0$, $|v_{\theta}| = 2U_{\infty}$ (Show as an ungraded homework)
- At $r = R$, $v_r = 0$ and $v_{\theta} = -2U_{\infty} \sin \theta$ (Show as an ungraded homework)

Ignore τ_f (friction)



$$N' = \int_0^c (P_l - P_u) dx$$

$$C_n = \frac{1}{c} \int_0^c (C_{P_l} - C_{P_u}) dx$$



AERODYNAMIC FORCES AND MOMENTS

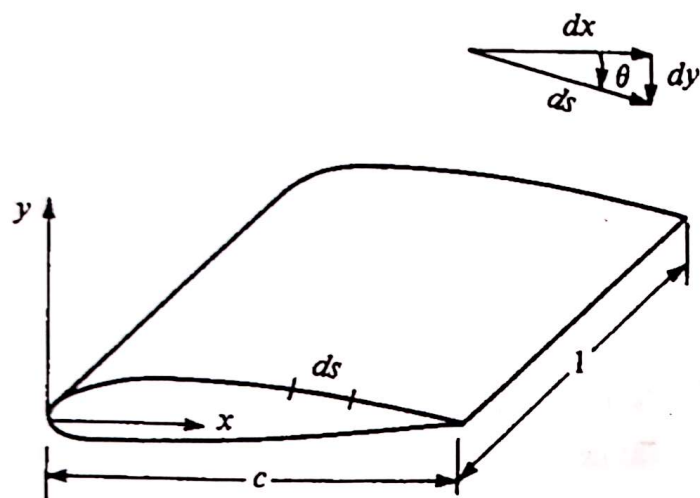
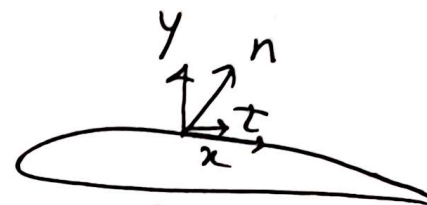
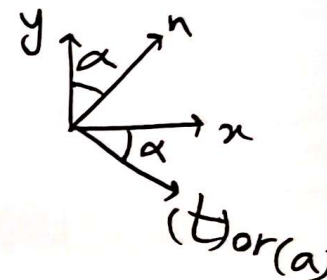


Figure 1.15 Geometrical relationship of differential lengths.

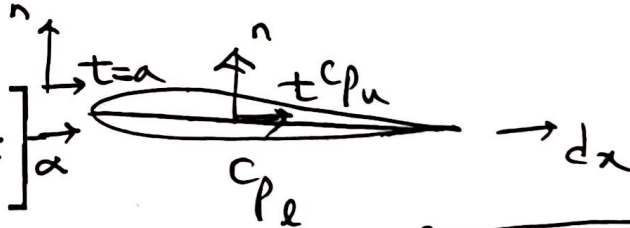


$$\begin{aligned} dx &= ds \cos \theta \\ dy &= -(ds \sin \theta) \\ S &= c(1) \end{aligned}$$



A.o.A

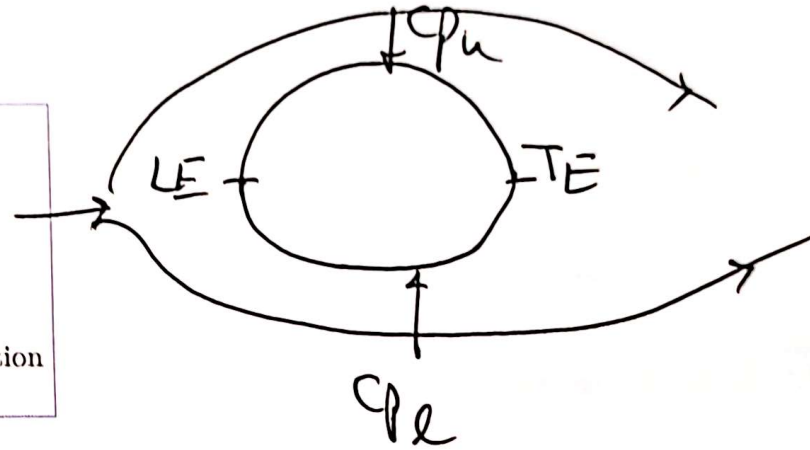
$$\begin{cases} \underline{c_l} = \underline{c_n} \cos \alpha - \underline{c_a} \sin \alpha \\ \underline{c_d} = \underline{c_n} \sin \alpha + \underline{c_a} \cos \alpha \end{cases}$$



$$\begin{aligned} c_n &= \frac{1}{c} \left[\int_0^c (C_{p,l} - C_{p,u}) dx + \int_0^c \left(c_{f,u} \frac{dy_u}{dx} + c_{f,l} \frac{dy_l}{dx} \right) dx \right] \\ c_a &= \frac{1}{c} \left[\int_0^c \left(C_{p,u} \frac{dy_u}{dx} - C_{p,l} \frac{dy_l}{dx} \right) dx + \int_0^c (c_{f,u} + c_{f,l}) dx \right] \quad \text{or} \quad c_a = \frac{1}{c} \int_0^c (c_{p,u} - c_{p,l}) dy \\ c_{m_{LE}} &= \frac{1}{c^2} \left[\int_0^c (C_{p,u} - C_{p,l}) x dx + \int_0^c \left(c_{f,u} \frac{dy_u}{dx} + c_{f,l} \frac{dy_l}{dx} \right) x dx \right. \\ &\quad \left. + \int_0^c \left(C_{p,u} \frac{dy_u}{dx} + c_{f,u} \right) y_u dx + \int_0^c \left(-C_{p,l} \frac{dy_l}{dx} + c_{f,l} \right) y_l dx \right] \end{aligned}$$

potential flow over a non-spinning cylinder

- $C_l = \frac{1}{c} \int_{le}^{te} (C_{pl} - C_{pu}) dy = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta = 0$. Can also be determined by observation of C_p distribution plot. There is top/bottom symmetry hence there is no net force in the y direction and thus no lift.
- $C_d = \frac{1}{c} \int_{le}^{te} (C_{pu} - C_{pl}) dx = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta = 0$. Can also be determined by observation of C_p distribution plot. There is left/right symmetry hence there is not net force in the x direction and thus no drag.



$$C_d = \frac{1}{c} \int_{le}^{te} (C_{pu} - C_{pl}) dy \quad (2.114)$$

$$y = R \sin \theta \rightarrow dy = R \cos \theta d\theta, \quad c = 2R \quad (2.115)$$

$$C_d = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta, \quad \text{Similarly } C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta \quad (2.116)$$

ential flow over a non-spinning cylinder

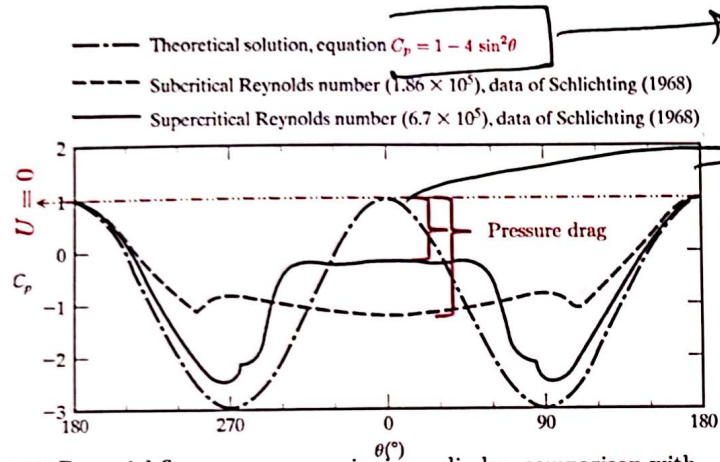
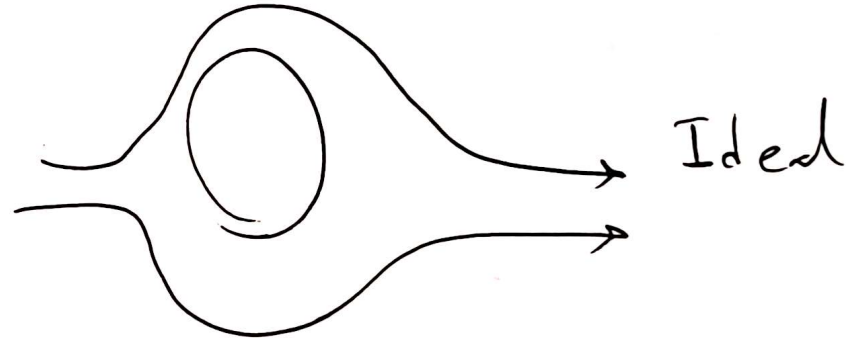
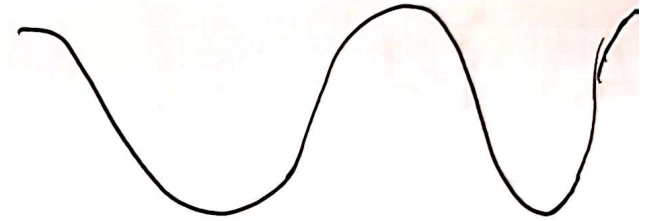


Fig. 2.22: Potential flow over a non-spinning cylinder, comparison with experiments, modified Fig. 3.17 from AE-BC

$$C_p = 1 - \left(\frac{v}{U_\infty} \right)^2 = 1 - \left(\frac{2U_\infty \sin \theta}{U_\infty} \right)^2 = 1 - 4 \sin^2 \theta$$

$C_{p \text{ theory from potential flow}}$



A hand-drawn diagram showing the streamlines of a viscous flow over a circular cylinder. The flow is asymmetric, with vortices (curved arrows) indicating flow separation and a wake region. The equation $D = \int (P_u - P_e) dy$ is written to the right.

ential flow over a spinning cylinder (11Sec. 3.15)

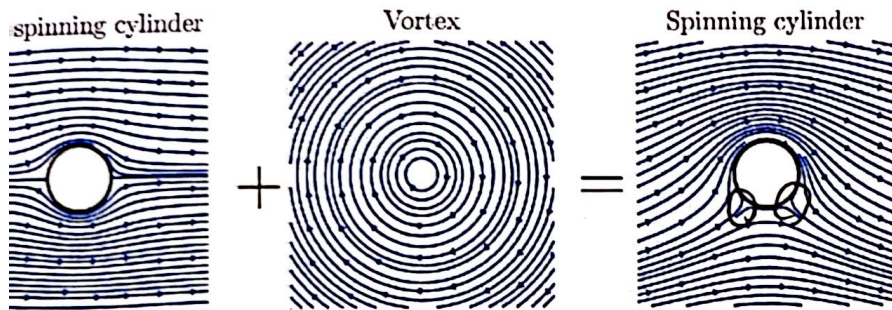


Figure 2.23: Potential flow over a spinning cylinder. Based on image from FA-JA

Uniform + Vortex

$$\psi(r, \theta) = \psi_{\text{free-stream}} + \psi_{\text{doublet}} + \psi_{\text{vortex}} \quad (2.117)$$

Doublet

$$= \psi_{\text{non-spinning cylinder}} + \psi_{\text{vortex}} \quad (2.118)$$

